massive class: people devoid of any economic, political or even artistic value, who contribute nothing to the prosperity, power and glory of society.

In September 2013 two Oxford researchers, Carl Benedikt Frey and Michael A. Osborne, published 'The Future of Employment', in which they surveyed the likelihood of different professions being taken over by computer algorithms within the next twenty years. The algorithm developed by Frey and Osborne to do the calculations estimated that 47 per cent of US jobs are at high risk. For example, there is a 99 per cent probability that by 2033 human telemarketers and insurance underwriters will lose their jobs to algorithms. There is a 98 per cent probability that the same will happen to sports referees, 97 per cent that it will happen to cashiers and 96 per cent to chefs. Waiters – 94 per cent. Paralegal assistants – 94 per cent. Tour guides – 91 per cent. Bakers – 89 per cent. Bus drivers – 89 per cent. Construction labourers – 88 per cent. Veterinary assistants – 86 per cent. Security guards – 84 per cent. Sailors – 83 per cent. Bartenders – 77 per cent. Archivists – 76 per cent. Carpenters – 72 per cent. Lifeguards – 67 per cent. And so forth. There are of course some safe jobs. The likelihood that computer algorithms will displace archaeologists by 2033 is only 0.7 per cent, because their job requires highly sophisticated types of pattern recognition, and doesn't produce huge profits. Hence it is improbable that corporations or government will make the necessary investment to automate archaeology within the next twenty years.[19]

Of course, by 2033 many new professions are likely to appear, for example, virtual-world designers. But such professions will probably require much more creativity and flexibility than your run-of-the-mill job, and it is unclear whether forty-year-old cashiers or insurance agents will be able to reinvent themselves as virtual-world designers (just try to imagine a virtual world created by an insurance agent!). And even if they do so, the pace of progress is such that within another decade they might have to reinvent themselves yet again. After all, algorithms might well outperform humans in designing virtual worlds too. The crucial problem isn't creating new jobs. The crucial problem is creating new jobs that humans perform better than algorithms.[20]